

Elevated Rates of High-Risk HPV+ DNA in IBD Patients

Introduction: IBD patients, especially Ulcerative Colitis patients, have significantly increased rates of anal cancer. However, no large-scale study has analyzed HPV rates of IBD patients, nor diagnosis rates of anal High-Risk HPV in relation to anal cancer. We did a retrospective analysis of the total IBD patient population in Epic Cosmos, an international dataset with roughly 2.4 million IBD patients (1.2 million Crohn's and UC patients each). We hypothesized that high risk HPV+ DNA on anal sampling or a history of perianal disease are risk factors that contribute to higher rates of anal cancer within IBD Populations.

Methods: We compared Crohn's and UC patients with an active diagnosis between 6/1990 to 6/2026. We looked at patients in those groups who had an ICD-10 diagnosis of High-Risk HPV+ DNA in anal tissue ("HR-HPV+ DNA") and patients with a diagnosis of Anal Cancer. We then analyzed for anal cancer patients stratified by theorized causes - history of perianal disease within the prior 5 years, history of azathioprine use after IBD diagnosis, and history of HR-HPV+ DNA within the prior 5 years. We performed chi-square analyses to look for dependence, and assumed a 95% Confidence interval when analysing for significance.

Results: Results are presented in the table below.

IBD Type	% with Anal HR-HPV+ DNA	% with Anal Cancer	% with Anal HR-HPV+ DNA, then Anal cancer within 5 years		% with Perianal Disease, then anal cancer within 5 years
Non-IBD	0.0075%	0.04%	0.003%		0.003%
Crohn's	0.016%	0.21%	0.008%		0.046%
Ulcerative Colitis	0.027%	0.27%	0.013%		0.04%

Compared with the general population, patients with IBD had a roughly 5-fold higher prevalence of anal cancer in Crohn's disease and 7-fold higher prevalence in ulcerative colitis.

Of the roughly 2.4 million IBD patients, 516 were found to have anal HR-HPV+ DNA. Compared with the general population, anal HR-HPV+ DNA prevalence was approximately 2.1-fold higher in Crohn's disease and 3.6-fold higher in ulcerative colitis. Chi Square analysis rejected independence with a $p < 0.001$, supporting a significant association between group status and outcome frequency.

Among patients with anal HR-HPV+ DNA, anal cancer diagnosis within 5 years was also more frequent in IBD groups: approximately 2.7-fold higher in Crohn's disease and 4.3-fold higher in ulcerative colitis compared with the general population. Similarly, among patients

with perianal disease, anal cancer diagnosis within 5 years was approximately 15-fold higher in Crohn's disease and 13-fold higher in ulcerative colitis.

Overall, ulcerative colitis showed the highest prevalence of anal cancer and anal HR-HPV+ DNA, while perianal disease was more strongly associated in Crohn's disease. Among the studied risk factors, perianal disease showed the largest relative increase in subsequent anal cancer diagnosis.

Conclusion: Compared with the general population, patients with IBD had significantly higher observed prevalence of anal cancer and anal HR-HPV+ DNA. Prior perianal disease was associated with an approximately 13- to 15-fold higher frequency of anal cancer diagnosis within 5 years. These findings may support further study of targeted HPV vaccination and screening strategies in IBD patients, particularly those with a history of perianal disease.